I underestimated AI capabilities (again)

Revisiting a prediction ten months early

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On Jan 14th, I made predictions about AI progress in 2026. My forecasts for engineering capabilities already feel much too conservative.

In my view, METR (where I now work) has some of the hardest and highest-software engineering and ML engineering benchmarks out there, and the m framework for making benchmark performance intuitive: we measure a tas difficulty by the amount of time a human expert would take to complete it ("time horizon"). [1]

When I made my forecasts last month, the model with the longest measure horizon on METR's suite of software engineering tasks was Claude Opus 4. succeed around half the time at software tasks that would take a human sof engineer about five hours. [2] Time horizons on software tasks had been doub less than twice a year from 2019 through 2025, which would have implied th the-art 50% time horizon should be somewhat less than 20 hours by the end But there was ambiguity about whether the more recent doubling time was the long-run trend, so I bumped that up to 24 hours for my median guess. [4] percentile was around 15 hours and my 80th percentile was around 40 hour

Now, Opus 4.6 (released only 2.5 months after Opus 4.5) was estimated to ha time horizon of ~12 hours. [5] I don't take the specific number literally — the fewer very-long tasks than medium and short tasks, and the long tasks mor guesstimated (rather than measured) human completion times, so time hori

estimates for the latest models are a lot noisier than they were in 2025. And benchmark underlying the time horizon graph is nearly saturated, which ca confidence intervals to blow up: the 95% CI is 5.3 hours to 66 hours. [6] It's re discriminate between different capability levels at the current range.

But at the end of the day, that dataset had 19 software engineering tasks est take humans longer than 8 hours, and Opus 4.6 was able to solve 14 of them some of the time (and it reliably nailed four of them). [8] And beyond just this suite, we've seen examples of AI agents doing certain very well-specified so tasks like writing a browser or C compiler, or porting a giant game, that wo humans many weeks or months to do on their own — not perfectly, but bet most people expected and better than a naive reading of the agents' measur horizon would have suggested.

And this happened in February. It's no longer very plausible that after ten w months of additional progress at the recent blistering pace, [9] AI agents wou struggle half the time at 24 hour tasks.

I wish them the best, but I think my colleagues on the capability evaluation METR might struggle to create new software tasks from a similar distributi of measuring AI agents' true time horizons through the end of the year. If w measure this, I'd guess that by the end of the year, AI agents will have a tim of over 100 hours on the sorts of software tasks in METR's suite (which are precisely specified — on certain extremely well-specified software tasks lik examples above, agents seem to *already* have a time horizon of more than a hours).

And once you're talking about multiple full-time-equivalent *weeks* of work, the whole concept of "time horizon" starts to break down.

It's nearly impossible to subdivide a typical one hour task (e.g., debugging c test) into smaller pieces that multiple people can work on in parallel. It wou

very well if you had to farm out writing this print statement or reading that message or tweaking this line of code to different people — the right actior depends intimately on everything that came before it and the precise state as a whole, you have to hold the whole context in your mind as you take eac they won't cohere in the right way.

It's somewhat easier to decompose an eight hour task (e.g., writing a simple game) into smaller components, but those components are constantly bleed each other in ways that make clean handoffs hard. When you're implementi game logic, you realize it needs to know something about how the graphics rendered. When you're handling user input, you find yourself tweaking the The fastest way to do it is probably one person knocking it out in a day, mal hundred small decisions fluidly as they go.

But it's actually pretty feasible to break down a *month*-long task into smalle fact, you may start *benefiting* from some explicit decomposition — it might to write a design doc laying out how the pieces fit together, or break the wo tickets so you don't lose track of what's done and what's left. And while it n one person working alone a month to complete, the fastest way to get it dor to have different people work on different pieces like the checkout flow or t inventory management panel in parallel.

And of course, tasks that take multiple full-time-equivalent *years* of work ne can and should be broken down into smaller milestones, and parallelized ac multiple teammates. Human work appears to get more and more decompos longer and longer it gets.

In other words, very few tasks feel *intrinsically* like year-long tasks, the way one bash command feels like an intrinsically one-second task, or debugging simple bug feels intrinsically like a one-hour task. Maybe a mathematician their head against a hard conjecture for a year before finally making a break a "real" year-long task? But most many-person-month software projects in

world sort of feel like they might be a bunch of few-week tasks in a trenchc way that a hundred-question elementary school math test is really 100 thirty tasks in a trenchcoat.

If an extreme version of that is true, then once AI agents can consistently d hour tasks, they should be able to make continuous progress on projects of scale. Maybe manager AIs can spend their work week figuring out how to f current project goal to line-worker AIs, line-workers can execute on their i piece, and all the AIs can maintain good enough records that no individual needs to build up holistic, long-term state on the whole project.

I think this *probably* won't fully work. Even projects with lots of formalized tracking still benefit a *lot* from everyone involved intuitively appreciating th picture in a way that isn't fully captured in Jira tickets and Asana tasks. Dec a 6 month project so cleanly and precisely that it can be executed by a team with no such holistic context might itself be, say, a 2 month task.

But it might work surprisingly well for a surprisingly large class of software AI agents are a lot cheaper and a *lot* more patient than humans, so it could l to get them to do far, *far* more task-tracking, documentation, and other proj management than human teams ever do. People have already started aggres experimenting with scaffolding for orchestrating agent teams. It's not clear will go over the next several months.

This is why my colleague Tom proposed that the calendar time it takes a lar humans to do a task might be a better proxy for "intrinsic difficulty" than th takes one human working alone. So far, the "team time" and "solo time" hav very similar in the METR task suite, since it has ranged from 1 second tasks 20 hour tasks. But we're entering the regime where these numbers could ra diverge. If Tom's conjecture is true, the "solo time" metric should start goin exponential about now…which makes it very hard to bound software engine capabilities by the end of the year.

In my predictions last month, my probability that AI R&D would be *fully* au the end of the year — AIs taking care of all the research ideation and imple no humans necessary [10] — was 10%. After I published that piece, I heard fro others in the AI forecasting space (including those I generally think of as m on AI timelines than I am) that it seemed a bit high. But now ten percent fe in the right ballpark again.

Fully automating AI R&D still seems like a tall order. Even fully automating engineering seems like it requires an aggressive read of the evidence, and A not just software engineering — it seems like automating it would require a amount of progress on "research judgment" and "creativity" and other ephe skills that AI systems still appear to be worse at than human researchers. I t lot more likely in the coming three or five years than this year.

But for the first time, I don't see any solid trend we can extrapolate to say it happen soon. [11] AI R&D really could be automated *this year*.

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1 It takes a bit of taste and judgment to decide what a task's "time horizon" is, and possible to game the metric to make it meaningless. Consider the "task" of takin math test consisting of 10,000 easy elementary school word problems — this is c 10,000 thirty-second tasks, rather than one 83-hour task. Or to take an example s my colleague Tom Cunningham, consider the task "Count how many times a ho mentioned in Anna Karenina." This might take a single human 10 hours, but it's parallelizable: a team of 300 people could each take one page and do the task in the METR task suite, "human time to complete" works as a good proxy for "intr difficulty because the tasks are constructed to be hard to easily decompose and Within each task, every piece depends on every other, and you benefit from keep whole context in mind. More on that later in the post.

2 Actually, at the time, METR was measuring models on its original time horizon 1.0), and the precise central estimate for Opus 4.5 was ~4h48min on that distribu since then, METR has released an updated suite with more tasks (TH 1.1) which models' scores to shift slightly; the measured time horizon for Opus 4.5 on TH 1 ~5h20min.

3 The 2019-2025 doubling time calculated in the original paper was 212 days, or 0. That means that if the time horizon at the beginning of January was 5 hours, the horizon at the end of the year should be 5 * exp(ln(2) * 1/ 0.58) ~= 16.4 hours. Whe doing mental math, I was approximating the 7 month doubling time as two doul 5 * 2 * 2 ~= 20 hours. The rule of thumb was a bit more aggressive than the strict extrapolation, but in fact, I just realized while writing this post that the mental extrapolation was too *conservative* in a different way — I should have dated the horizon to Nov 24 (Opus 4.5's release date), not the beginning of January, meani have added an extra month to all these extrapolations.

4 I didn't actually do this math at the time, but the original paper calculated a dou of 118 days or 0.32 years since 2024, and if you assume that doubling time was c the time horizon by EOY 2026 should have been 5 * exp(ln(2) * 1/0.32) ~= 43 hours I took this faster doubling time pretty seriously at this time, this suggests my m especially my 80th percentile should have been higher to begin with. I would pr been better served by the heuristic of using just the previous year (rather than th seven years) to forecast the next year.

5 Originally, METR estimated Opus 4.6 to have a 50% time horizon of ~14.5 hours 2026. We corrected a bug in our modeling on March 3, 2026 and this reduced its horizon estimate to ~12 hours. Note that this measurement was done on Time H the new task suite released on Jan 29, 2026.

6 This is something I appreciate about the time horizon construct. For standard b confidence intervals around a point get *narrower* as the benchmark approaches s a model gets 50% accuracy, there's more room for error in either direction than i accuracy. But since the time horizon metric could get infinitely long, the confid intervals can be constructed so uncertainty gets *wider* rather than narrower as th hardest tasks are saturated. Epistemically, it's appropriate for your uncertainty a capabilities to blow up when you no longer have tasks that the models can't com

7 In the original time horizon paper, METR measured human completion times fo the tasks in the dataset by actually having humans do them (148 out of 169). In th suite, only 5 of the 19 tasks longer than 8 hours have measured human baselines others are estimated.

8 Agents are usually run on the same task 6 separate times. The same agent doesn approach the same task the same way each time — it sometimes gets lucky or un four of the 19 hard tasks, Claude Opus 4.6 solved it successfully in all six runs; fo ten, it solved it successfully in at least one of the six runs.

9 If you look just at the year 2025, agent time horizons doubled every *~3.5 months*, months as in the long-run trend or even every ~4 months as in the 2024-2025 tre didn't factor this into my forecasts; it wasn't salient to me compared to the rule absorbed of "two doublings a year." If I had used this to extrapolate from Opus 4 would have suggested time horizons by the end of the year should be 5 * exp(ln(2 54 hours by the end of the year.

10 Specifically, my operationalization was that firing all human members of techni (research leads, engineers, everyone) would only slow down progress by 25%. No past, of course, firing every single human would cause progress to completely ha

11 A concrete example of this is the kind of argument that my colleague Nikola ma Nov 2025, when the state-of-the-art time horizon was 2 hours; this kind of argu be much shakier made today (less than four months later!).

← Previous

Discussion about this post

Comments Restacks

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Will Kiely 5d

Even if 50% Success is going super-exponential now, wouldn't you expect 80% Suc go exponential before full automation of AI R&D happens?

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12 replies

Jacob Asmuth 5d

Agreed with everything written in the comments to you so far. Lots of useful feedba iterate on!

1. We already measure 80%, so simply stop reporting 50% as it is clearly becoming therefore less useful.

2. 80% seems to lag 50% by nearly a full year, so that buys you a LOT of time for yo build better tasks.

3. Clearly AI agents need much higher success rates than 80%. Investigate 95% su measurements in the future for further savings as your tasks begin to become satur

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